Medical and Surgical Imaging Challenge – Sequential Versus Spiral CT

This coursework is on the analysis and design of a medical imaging task. You should extend what you have learned on the medical and surgical imaging module, summarising your findings in a short presentation.

Starting with the 'CT projections' data provided in the X-ray/CT Classwork, carry out a qualitative and quantitative comparison of axial/sequential muti-slice CT acquisition and helical/spiral CT acquisition.

The data that you are provided with consists of a series of projections of a computational CT phantom. Let us assume that the CT system is a multi-detector device where the detector has 10 rows and the data was acquired with an axial/sequential multi-slice CT system where the table was paused 19 times to complete the full CT scan. You can ignore any beam fan effects and assume that the X-ray direction was exactly perpendicular to the detector element surface.

For the first spiral CT scan, assume that the table moves by the width of the CT detector during one complete detector rotation (i.e. the pitch is equal to 1.0). To obtain projections through path positions that are not available in the original projection dataset, you can carry out an interpolation from neighbouring positions. For the second spiral CT scan, assume a pitch equal to 2.0.

Reconstruct the CT data for the 110th slice (sequential/spiral 1.0/spiral 2.0) and carry out a qualitative and quantitative analysis of the data. Also, discuss the advantages and disadvantages of spiral CT.

Your presentation should clearly build on and link with the material covered during the course. You should discuss the effects of any assumptions you have made as well as providing clinical context where possible.

You will be assessed via an in-person presentation (e.g. using Powerpoint) including questions from the assessors, during the week of 3rd February. Keep the descriptions of your work to be precise and succinct. You can assume that the assessors have a high level of knowledge of standard medical imaging techniques and this background should **not** be covered during your presentation. You will be expected to submit a version of your presentation together with copies of any code you have developed via a zip file on Blackboard by 4 pm on Friday 31st January.

Marks will be awarded as follows:

- Clear links to the lecture course (20%)
- Accuracy of presentation (30%)
- Clarity of explanation (30%)
- Answers to questions (20%)

Further information about the assessment and submission can be found on the BB site and in the discussion forum.